Linguistics - Structural Correspondence with the Five-Stage Model

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Linguistics

Structural correspondence survey with the five-stage model (Ba, Nami, En, Uzu, Taba)

Survey Overview

- **Scope**: 10 major theories in linguistics
- **Research question**: Do theories in linguistics structurally correspond to the five-stage model?
- **Results**: Strong correspondence 6, Partial correspondence 2, Conditional correspondence 2

Overview of the Five-Stage Model

Stage	Definition
Ba (Field)	An undifferentiated state. Initial conditions where neither direction nor structure has yet been determined
Nami (Wave)	A stage of exploration where multiple directions diverge and compete
En (Edge)	A state of tension where opposing elements coexist without converging to either side. A place where elements meet at boundaries, influence each other, and relationships emerge
Uzu (Vortex)	A stage where new coherence (order) spontaneously arises from within tension
Taba (Bundle)	A stage where form is established and stabilizes as a reusable structure

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Figure 1: Domain — Research Overview Infographic

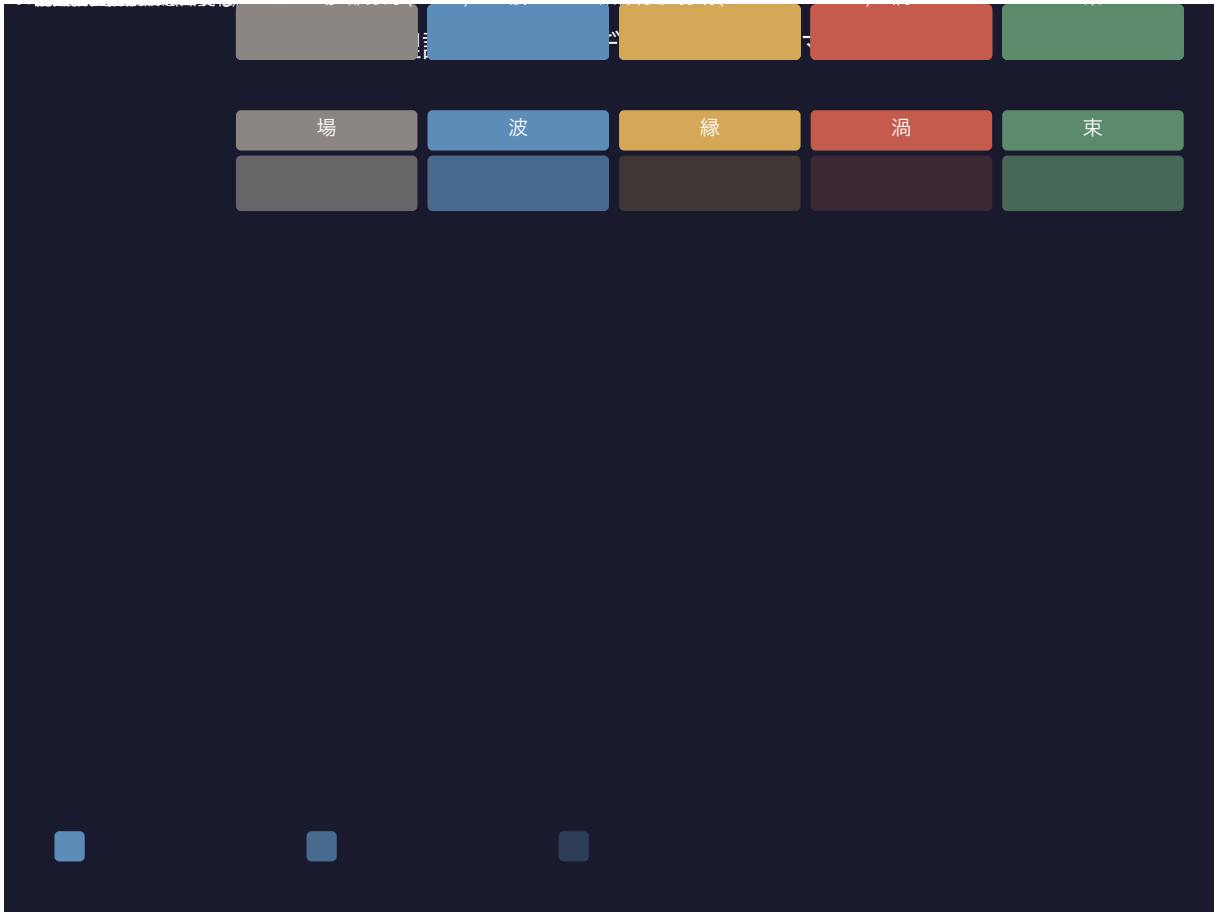

Figure 2: Domain — Theory x 5-Stage Correspondence Matrix

Overall Structural Correspondence

#	Theory/Concept	Proponent	Primary Corresponding Stages	Assessment
1	Grammaticalization	Hopper & Traugott (2003)	Ba→Nami→En→Uzu→Taba	Strong correspondence
2	Diffusion model of sound change	Labov (1994, 2001)	Ba→Nami→En→Uzu→Taba	Strong correspondence
3	Creole formation	Thomason & Kaufman (1988)	Ba→Nami→En→Uzu→Taba	Conditional correspondence
4	Critical period hypothesis	Lenneberg (1967)	Ba→Nami→En→Uzu→Taba	Partial correspondence
5	Nicaraguan Sign Language	Senghas et al. (2004)	Ba→Nami→En→Uzu→Taba	Strong correspondence
6	Emergence of constructions	Tomasello (2003), Goldberg (1995)	Ba→Nami→En→Uzu→Taba	Strong correspondence
7	Pragmatic inference and semantic change	Grice (1975), Sperber & Wilson (1986)	Nami→En→Uzu	Partial correspondence
8	Linguistic relativity	Slobin (1996), Boroditsky (2001)	Taba→Ba	Conditional correspondence
9	Optimality Theory	Prince & Smolensky (1993/2004)	Ba→Nami→En→Uzu→Taba	Strong correspondence
10	Iterated learning model	Kirby et al. (2014)	Ba→Nami→En→Uzu→Taba	Strong correspondence

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Key Entry 1: The Emergence and Systematization of Nicaraguan Sign Language (Senghas et al., 2004)

- Nicaraguan Sign Language (NSL) is a sign language that newly emerged in the late 1970s in a school for the deaf in Nicaragua. It is known for the fact that a grammatical system formed spontaneously through the interactions of children in the absence of a language model. Senghas et al. (2004, *Science*) observed in real time the process by which grammatical structure became increasingly complex with successive generations, by comparing cohorts of different enrollment years.
- **As fact**: The first generation of children brought together their own uncoordinated individual means of expression (home signs). According to the analysis by Senghas et al. (2004), subsequent generations showed the ability to decompose events into basic elements and reconstruct them hierarchically. Spatial modulation differentiated into the two functions of "who" and "where" (Senghas, 2011), and the process of creation and convergence in temporal expression has also been confirmed through multi-cohort comparison (Kocab et al., 2016).
- **As reading**: Here we read the process by which linguistic structure is spontaneously organized through intergenerational transmission from zero. The level of similarity is process. Despite the fact that individual speakers did not intend to "create grammar," the core grammatical features of discreteness, combinatoriality, and hierarchy emerged through generational turnover. Particularly noteworthy is that transitions between stages are made visible on a social timeline through cohort comparison.
- **As interpretation**: The coexistence of uncoordinated expressive resources corresponds to Ba; initial variation and competition corresponds to Nami; functional differentiation and the assignment of devices corresponds to En; the establishment of a system with discreteness and combinatoriality corresponds to Uzu; and the norms that remain through intergenerational propagation and convergence within the community correspond to Taba. NSL is one of the most direct cases in which the entire process of the five stages can be observed in real time. The fact that grammar emerges

without the intention of individual speakers constitutes concrete support for the claim that Uzu in the five-stage model is "spontaneous order formation."

Key Entry 2: The Chain of Grammaticalization (Hopper & Traugott, 2003)

- Grammaticalization is the irreversible process by which lexical items shift to grammatical functions. For example, English "going to" originally meant movement in the sense of "going toward ~," but eventually changed to the future-tense grammatical marker "gonna." Hopper & Traugott (2003) showed that this process can be described as a unidirectional change (cline) from content word → grammatical word → affix → zero.
- **As fact**: Grammaticalization exhibits cross-linguistically isomorphic pathways. The pathway by which words denoting body parts become spatial prepositions and then temporal expressions has been repeatedly confirmed even in languages that are genealogically unrelated (Heine et al., 1991). The driving force of this change lies in pragmatic inference — conversational implicatures that are repeated and become conventionalized (Traugott, 1989).
- **As reading**: Here we read the chain structure in which lexical items irreversibly shift to grammar. The level of similarity is structure. Noteworthy are the points that the change is unidirectional and that isomorphic pathways are observed in genealogically unrelated languages. The former suggests that there is direction to transitions between stages; the latter suggests the existence of structural patterns not specific to any particular language.
- **As interpretation**: The polysemy that a lexical item has before grammaticalization corresponds to Ba; the stage in which new usages arise and compete through pragmatic inference corresponds to Nami; the phase in which old and new usages coexist and functions are re-analyzed corresponds to En; the stage in which a coherence is established as a grammatical function corresponds to Uzu; and the stage in which it is consolidated as part of the grammatical system and becomes the foundation for subsequent changes corresponds to Taba. The observation of cross-linguistically isomorphic pathways is consistent with the hypothesis that the structure of the five stages is not specific to any particular domain.

Key Entry 3: The Emergence of Constructions — Usage-Based Model (Tomasello, 2003; Goldberg, 1995)

- The usage-based approach is the theory that linguistic structure emerges from individual instances of use. Rather than assuming innate knowledge specific to the linguistic domain, structure is derived from general cognitive abilities: statistical learning, chunking, categorization, and analogy. Tomasello (2003) analyzed children's syntactic development, and Goldberg (1995) analyzed constructions as units that independently carry meaning as form-meaning pairings.
- **As fact**: Children initially produce constructions in forms tied to specific vocabulary. For example, they express change of location only in sentences containing "put," and do not apply the same construction to other verbs. Through repetition and generalization, they eventually arrive at productive use (Tomasello, 2003). Bybee & Beckner (2010) showed that frequency of use drives pattern formation, that frequent combinations are stored as chunks, and that chunks are generalized into categories.
- **As reading**: Here we read the process in which abstract construction patterns gradually emerge from the accumulation of concrete instances of use. The level of similarity is mechanism. The causal chain in which frequency bias drives chunking, and chunking drives categorization, concretely shows the mechanisms of transitions between stages.
- **As interpretation**: The accumulation of concrete instances of use corresponds to Ba; frequency bias and distributional gradients correspond to Nami; relational mapping through chunking, categorization, and analogy corresponds to En; the establishment of a construction (a form-meaning pairing) corresponds to Uzu; and the acquisition of generalization and productivity as a construction network corresponds to Taba. This theory, in not assuming linguistic-domain-specific innate knowledge, is consistent with the view that the structure of the five stages is not limited to language.

Cross-Cutting Patterns

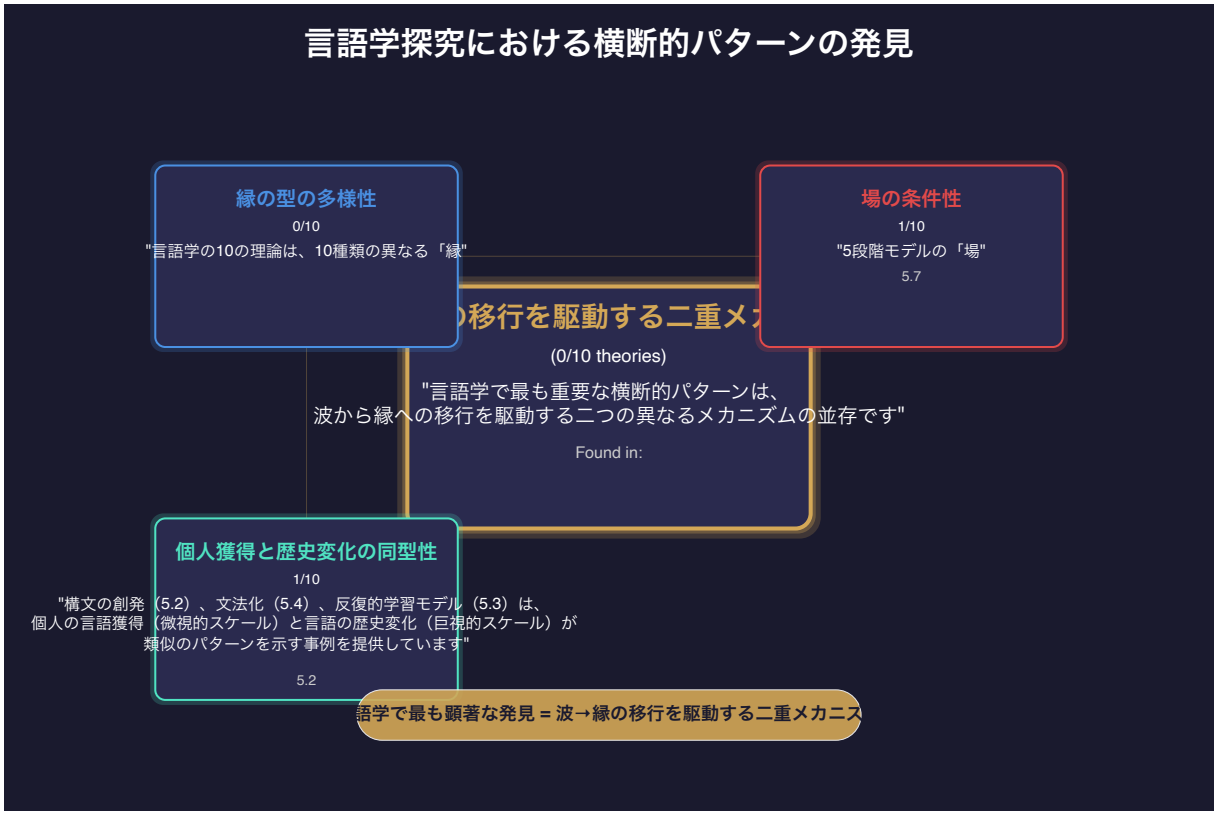

Figure 3: Domain — Cross-Cutting Patterns Diagram

- The most striking pattern across linguistics is the **diversity of implementations of En**
- Second, the point that **many language changes are unintended creations** is common across multiple theories
- Third, **isomorphism across scales** is prominent
- Fourth, it has become clear that **the mechanism of Ba formation is not uniform**

Unresolved Questions

- Whether grammaticalization (historical change) and the emergence of constructions (individual acquisition) are different-scale versions of the same mechanism remains without conclusion. It is necessary to verify whether what the two share is structure or merely label similarity.
- Whether the emergence of grammar in Nicaraguan Sign Language should be understood as "unintended collaboration" or "implicit selection pressure" is divided. This distinction relates to the question of the nature of Uzu — whether it is completely spontaneous order formation or includes direction by environmental pressure.
- The findings of linguistic relativity (linguistic systems constrain cognitive frameworks) are the only group of theories describing the influence relationship from Taba back to Ba, but they are themselves not theories of process but descriptions of state constraint relationships. The relationship with the hypothesis that the "Taba→Ba return" in the five-stage model is spiral requires further clarification.
- In creole formation (conditional correspondence), the extent to which diversity across cases (power relations, population ratios, structural distance of input languages) and structural common patterns can be separated remains a challenge.

Conclusion

- This survey confirmed that linguistics is a domain with strong overall structural similarity to the five-stage model
- In particular, the emergence and systematization of Nicaraguan Sign Language, the iterated learning model, and the emergence of constructions are theories that concretely illuminate the entire process of the five stages
- The greatest contributions from linguistics are two